

Rotation-Equivariant Variational Autoencoders for Histopathology Representation Learning and Downstream Classification

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Abstract—Histopathology patches have no canonical upright orientation, and the diagnostic and tissue labels considered here are unchanged by global in-plane rotation. Expert annotations remain costly and substantially less abundant than unlabeled image regions. This study evaluates whether continuous rotation equivariance improves the representations learned by a variational autoencoder (VAE) and whether those representations are useful for subsequent classification. A conventional convolutional VAE and a continuous-SO(2)-equivariant VAE were trained on the same 300,000 ovarian-cancer patches from 322 whole-slide images (WSIs), using the same denoising objective, latent shape and budget of 60,000 updates. Their checkpoints were then frozen. Evaluation used an independent cohort of 152 WSIs divided into 106 training, 23 validation and 23 test slides for supervised tasks. The test set contained 67,138 patches for reconstruction, 23 complete bags for five-class WSI diagnosis and 31,572 annotated patches for three-class tissue recognition. The conventional VAE obtained a lower mean absolute reconstruction error, although the paired WSI-bootstrap interval for the primary difference included zero. For WSI diagnosis, the equivariant representation obtained higher macro-F1 (0.483 versus 0.389) and accuracy (0.609 versus 0.522), but the corresponding intervals did not establish a difference. For tissue recognition, it produced higher macro-F1 at four of five label budgets; only the 500-label-per-class comparison had a simultaneous 95% interval excluding zero, and the largest budget favored the conventional representation. The primary analysis did not establish a reconstruction difference between the two implemented VAEs or a general classification advantage. The WSI macro-F1 interval and four of the five simultaneous tissue macro-F1 intervals included zero.

Index Terms—histopathology, variational autoencoder, rotation equivariance, SO(2), whole-slide image classification, representation learning

I. INTRODUCTION

Digital histopathology converts glass slides into whole-slide images whose dimensions commonly reach gigapixel scale. These images contain many local tissue regions, but their diagnostic labels and detailed annotations require specialist review. Representation learning from unlabeled patches offers a way to use this image volume before fitting a smaller supervised model [1]–[3].

Histopathological images also differ from many natural-image datasets because local tissue structures do not have a fixed upright orientation. A tissue pattern can appear at any in-plane angle without changing its biological meaning.

Conventional convolutional networks account for this variation mainly through rotated examples or by learning separate filters for different orientations. Rotation-equivariant networks instead constrain intermediate representations to transform according to the same rotation applied to the input [4]–[6]. This construction is relevant to histopathology, where rotation-equivariant classifiers have previously been evaluated for patch-level prediction [7].

An equivariant architecture does not, however, ensure that a trained model will reconstruct images more accurately or provide a better representation for every supervised task. The effect must be measured under a common data and training protocol. In this study, two denoising VAEs [8] are compared: a conventional residual convolutional model and a model whose convolutional kernels are constrained to be equivariant to the continuous planar rotation group SO(2). Both models produce a spatial Gaussian latent representation and are frozen before any diagnostic or tissue label is used.

The experiment addresses two questions:

- 1) Does the rotation-equivariant VAE improve reconstruction of unseen histopathology patches relative to the conventional VAE?
- 2) Does the frozen equivariant representation improve classification at WSI level or tissue-patch level under the same supervised protocol?

The work consists of an experimental characterization of the two VAEs and an evaluation of their contribution to two downstream classification tasks. The comparison preserves the image populations, spatial latent dimensions, reconstruction objective, optimization budget and supervised data assignments. Results are reported with uncertainty at WSI level because patches from one slide are not independent biological observations.

II. RELATED WORK

A. Representation Learning in Histopathology

Whole-slide images cannot generally be processed at native resolution in one forward pass. A common approach divides each WSI into patches, learns a local representation, and aggregates patch features for slide-level prediction. Tellez *et al.* used neural image compression to learn patch codes before

gigapixel prediction [3]. Attention-based multiple instance learning (MIL) provides a related supervised formulation in which a WSI is treated as a bag of instances and only the bag label is required [9]. Subsequent work has applied weakly supervised MIL to large histopathology cohorts [10].

The present study follows this two-stage organization. The VAE learns from unlabeled patches. Its posterior mean is then used either as a patch-level input for tissue recognition or as the instance representation of a WSI bag. The VAE weights remain fixed in both cases.

B. Rotation Equivariance

For an input transformation T_g , a representation map E is equivariant when there is a corresponding latent action ρ_g such that

$$E(T_g \mathbf{x}) = \rho_g E(\mathbf{x}). \quad (1)$$

Group-equivariant convolutional networks construct layers that satisfy this relation for a selected transformation group [4]. Steerable convolutions extend the construction to continuous rotations by restricting the kernel space according to the representations carried by the input and output fields [5], [6]. When the target function follows the same symmetry, this restriction can reduce the effective hypothesis space [11]. It also changes parameter count, computation and optimization, so an empirical comparison does not isolate symmetry unless these factors are separately controlled.

III. MATERIALS AND METHODS

A. Experimental Design

The experiment had two stages. First, the conventional and equivariant VAEs were trained with unlabeled patches. Second, the final checkpoints were frozen and their deterministic posterior means were used for reconstruction and supervised classification. No downstream label was used to update either VAE.

Both VAEs received the same ordered patches, normalization, corruption policy, loss, number of updates and validation schedule. The supervised comparisons also used the same WSI assignments, patch coordinates, classifier initialization, epoch order and model-selection rule for the two representations. The experiment therefore compares two complete frozen representations under a common protocol; it does not estimate the isolated causal effect of equivariance.

B. Data Source and Image Units

The data were obtained from the UBC-OCEAN ovarian-cancer resource [12]. The diagnostic labels are clear cell carcinoma (CC), endometrioid carcinoma (EC), high-grade serous carcinoma (HGSC), low-grade serous carcinoma (LGSC), and mucinous carcinoma (MC). Tissue microarrays were excluded because their scale and layout differ from conventional WSIs.

Each WSI was represented by non-overlapping RGB patches of 256×256 pixels. The biological unit used to separate data populations was the WSI, not the patch. Patches

from one slide were never assigned to different train, validation or test subsets.

Figure 1 shows one WSI and two actual patches used in the VAE validation population. Both patches originate from WSI 59031 and retain their full-resolution coordinates. The thumbnail is used only to illustrate the relation between a slide and the local inputs; model inference uses the original 256×256 crops.

C. Patch Selection and Tissue Annotations

Foreground selection for reconstruction and WSI diagnosis was derived from the image rather than from human masks. Each distributed WSI thumbnail was converted to HSV, and Otsu's threshold was applied to the saturation channel [13]. A full-resolution grid cell was retained when more than 60% of its projected thumbnail area belonged to foreground. This procedure prevented the supplemental tissue annotations from determining the patches used for the primary reconstruction and diagnostic tasks.

Supplemental masks were used only to construct the tissue-recognition population. Red, green and blue mask regions were interpreted as tumor, stroma and necrosis. The masks are not exhaustive; an unpainted region cannot be treated as a negative class. A patch was therefore eligible for tissue classification only when at least 10% of its area was annotated and at least 95% of that annotated area belonged to one tissue class. The resulting task classifies high-purity annotated regions and does not constitute dense pixel-level segmentation of the complete WSI.

D. Data Populations and Partitions

Two non-overlapping WSI populations were used. VAE development used 322 WSIs for training and 39 for validation, represented by 300,000 and 30,000 patches, respectively. The later evaluation cohort contained 152 different WSIs and was fixed as 106 supervised-training, 23 validation and 23 test slides before VAE or classifier outputs on that cohort were inspected.

The class counts for train/validation/test were 23/5/5 CC, 24/6/6 EC, 40/8/8 HGSC, 11/2/2 LGSC and 8/2/2 MC. The assignment was shared by both downstream tasks. It used diagnosis and the availability of annotated tissue to maintain a feasible rare-tissue evaluation, but it did not use image pixels, VAE representations or model predictions. Figure 2 shows the complete experimental partition.

E. Conventional Variational Autoencoder

The conventional model was derived from a residual convolutional design informed by ResNet-18 [14], with the down-sampling schedule limited to preserve a spatial latent map. Its encoder used stages with 32, 48, 64 and 96 channels at spatial resolutions 256, 128, 64 and 32. Each stage contained two residual blocks. Fixed anti-aliased downsampling was applied between encoder stages. The decoder reversed the schedule through bilinear upsampling followed by convolution.

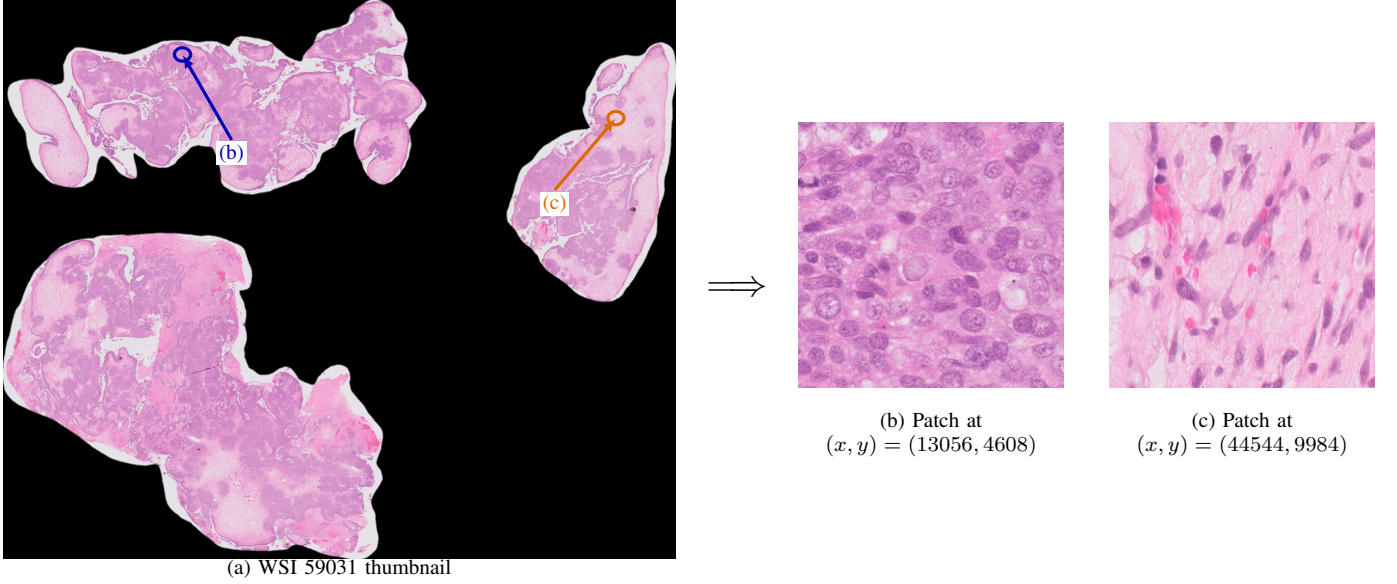


Fig. 1. Example of the whole-slide and patch data hierarchy. The markers in (a) show the full-resolution locations of the two archived 256×256 patches in (b) and (c) from WSI 59031 (HGSC). The thumbnail is shown for context and is not the image supplied to the VAE.

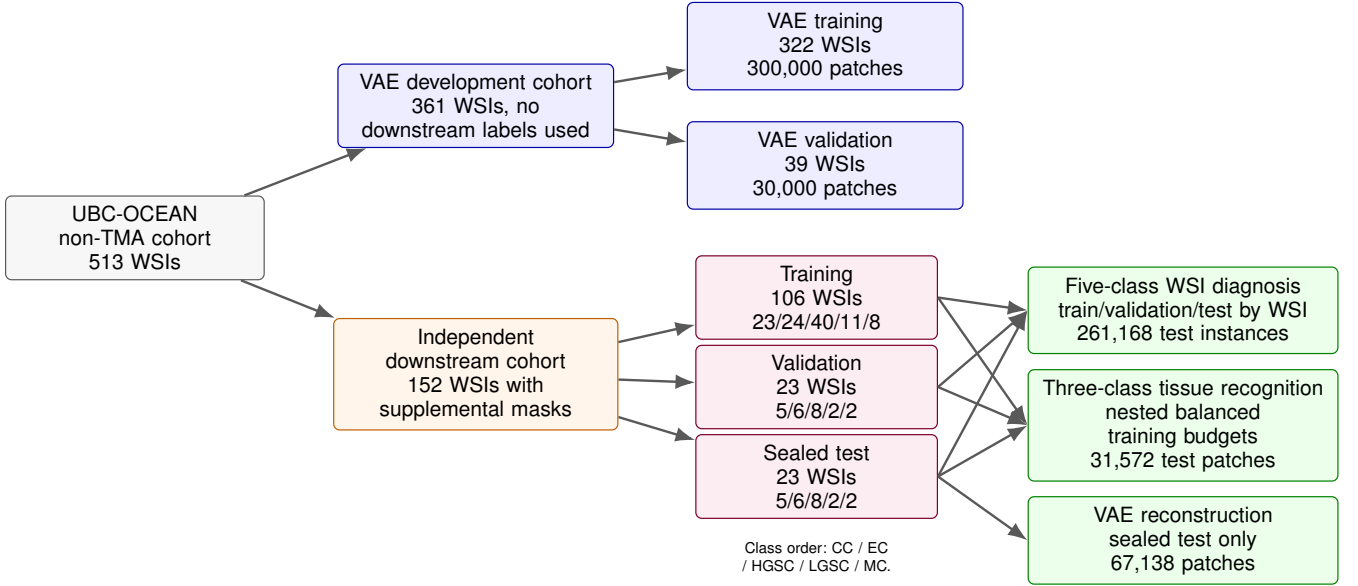


Fig. 2. Data populations and WSI-level partitions. The VAE-development and downstream cohorts are disjoint. All three supervised subsets are fixed at WSI level and shared by the conventional and equivariant representations. Numbers inside split boxes follow the order CC/EC/HGSC/LGSC/MC.

Ordinary learned convolutions, GroupNorm and learned scalar gates were used throughout the hidden layers. Separate 5×5 heads produced the posterior mean and log-variance

$$\mu, \log \sigma^2 \in \mathbb{R}^{16 \times 32 \times 32}. \quad (2)$$

The decoder received a sampled latent tensor with the same shape and returned a raw three-channel image without a final tanh operation. The conventional model contained 3,958,435 learned parameters.

F. Continuous-SO(2)-Equivariant Variational Autoencoder

The equivariant VAE used the same input, output, spatial schedule, number of residual blocks and posterior shape. Ordinary convolutional kernels were replaced by steerable kernels constrained for the continuous group $\text{SO}(2)$. Hidden feature maps combined scalar fields F_0 and first-order vector fields F_1 . The four encoder stages contained 16, 24, 32 and 48 copies of each field type. Field-aware normalization was used, and each F_1 pair shared a radial gate so that the nonlinearity did not introduce an orientation-dependent component.

The kernel basis was evaluated on fixed spatial grids and the learned parameters were the expansion coefficients of that

TABLE I
POPULATIONS USED FOR REPRESENTATION LEARNING AND EVALUATION.

Population	WSIs	Patches/instances	Role and sampling unit
VAE training	322	300,000	Unlabeled 256×256 patches used to optimize both VAEs.
VAE validation	39	30,000	Unlabeled patches used for scheduled validation; no downstream labels.
Supervised training	106	task dependent	WSI bags for diagnosis and nested balanced patch subsets for tissue recognition.
Supervised validation	23	task dependent	Independent checkpoint selection for each frozen representation and task.
Sealed reconstruction	23	67,138	Foreground patches; uncertainty aggregated by WSI.
Sealed WSI diagnosis	23	261,168	Complete selected foreground bags; the principal unit is one WSI.
Sealed tissue recognition	23	31,572	High-purity annotated patches: 21,796 tumor, 8,500 stroma and 1,276 necrosis.

basis. The final posterior heads produced 16 scalar fields, giving the same $16 \times 32 \times 32$ Gaussian latent shape as the conventional VAE. The equivariant model contained 1,180,035 learned parameters, or 29.8% of the conventional count. The kernel expansion nevertheless increased the estimated convolutional work, so the lower parameter count should not be interpreted as lower computational cost.

G. VAE Objective and Training

Both models were trained as denoising VAEs. With probability 0.3, the input patch was perturbed in hematoxylin/eosin/residual coordinates using color deconvolution [15], and Gaussian noise was added. The target remained the corresponding clean patch. RGB values were normalized to $[-1, 1]$.

For a clean target $\mathbf{x}$ and reconstruction $\hat{\mathbf{x}}$, the objective was

$$\mathcal{L} = \text{MAE}(\hat{\mathbf{x}}, \mathbf{x}) + 0.1 [1 - \text{SSIM}(\hat{\mathbf{x}}, \mathbf{x})] + \beta D_{\text{KL}}(q_{\phi}(\mathbf{z} | \tilde{\mathbf{x}}) \| \mathcal{N}(\mathbf{0}, \mathbf{I})), \quad (3)$$

where SSIM is the structural similarity index [16] and $\tilde{\mathbf{x}}$ denotes the corrupted input. The KL weight increased linearly from zero to 0.01 during the first 6,000 updates and remained fixed thereafter.

AdamW [17] used coefficients (0.9, 0.999), $\epsilon = 10^{-8}$, weight decay 10^{-5} and gradient-norm clipping at 1.0. The learning rate increased linearly for 600 updates to 10^{-3} and then decreased with a cosine schedule to 10^{-5} . Each model was trained for 60,000 successful updates with a global batch of 50 patches distributed over two NVIDIA T4 GPUs. Validation and checkpointing occurred every 3,000 updates. The comparison used the final update-60,000 checkpoint for both models rather than selecting different VAE checkpoints from the downstream results.

H. Frozen Representations

After VAE training, all encoder and decoder parameters were frozen. Downstream tasks used the posterior mean μ rather than a sampled latent. For each selected coordinate, the RGB patch was read once and passed through both encoders. The two FP32 latent records were stored in the same logical order, which made WSI membership, tissue labels and patch coordinates identical between representation branches.

I. Reconstruction Evaluation

The reconstruction test contained 67,138 foreground patches from the 23 test WSIs, with at most 3,000 systematically distributed coordinates per slide. MAE and mean squared error (MSE) were measured in the normalized $[-1, 1]$ domain. Peak signal-to-noise ratio (PSNR) and SSIM were measured after conversion to $[0, 1]$ and clamping. Mean MAE was the primary reconstruction endpoint. The other metrics describe complementary error properties.

J. WSI Diagnosis

The diagnostic task predicted the five ovarian-cancer subtypes. Every selected foreground patch was represented by its frozen posterior mean. A trainable convolutional patch encoder compressed the $16 \times 32 \times 32$ latent map. Two local attention blocks exchanged information within a radius-two spatial neighborhood, and a global stream summarized the complete WSI through learned sigmoid attention. A final classification layer produced five logits. The supervised model contained approximately 1.51 million parameters.

Training used one complete WSI bag per optimizer step, AdamW and class-weighted cross-entropy computed from the 106 training WSIs. Each representation branch started from the same classifier weights and received the same WSI order. The maximum was 150 epochs, validation was performed twice per epoch, and each branch independently retained its highest validation macro-F1 checkpoint. The primary test metric was macro-F1; balanced accuracy, accuracy, mean cross-entropy, classwise F1 and confusion matrices were secondary summaries.

K. Tissue Recognition and Label Budgets

The tissue task predicted tumor, stroma or necrosis from a single frozen latent map. The classifier contained three convolutional stages $16 \rightarrow 32 \rightarrow 64 \rightarrow 128$, GroupNorm, GELU activations [18], adaptive average pooling and a linear $128 \rightarrow 3$ output layer.

Training subsets contained 250, 500, 1,000, 2,500 and 5,671 labeled patches per class. They were class balanced, nested and selected without replacement; every smaller set was a subset of each larger one. WSI-stratified ordering ensured that the smallest subset did not collapse onto a single slide. Each normal/equivariant pair started from identical classifier

TABLE II
ARCHITECTURE COMPARISON.

Property	Conventional VAE	Continuous-SO(2) VAE
Hidden representation	Ordinary channels: 32, 48, 64, 96	Scalar/vector field copies: 16+16, 24+24, 32+32, 48+48
Learned convolutions	43 ordinary convolutions	43 steerable convolutions
Normalization/nonlinearity	GroupNorm and learned scalar gates	Field-aware normalization and scalar/radial gates
Posterior	$\mu, \log \sigma^2 \in \mathbb{R}^{16 \times 32 \times 32}$	Same spatial Gaussian posterior, represented by scalar fields
Learned parameters	3,958,435	1,180,035
Estimated convolutional work	3.65×10^{10} MAC/sample	1.598×10^{11} MAC/sample
Rotation constraint	None beyond translation-equivariant convolution	Continuous SO(2) kernel constraint

weights and received the same patch order. Models were optimized for at most 30 epochs, and the best validation macro-F1 checkpoint was retained. The test distribution was not balanced and contained 21,796 tumor, 8,500 stroma and 1,276 necrosis patches from 23 WSIs.

L. Statistical Analysis

All difference intervals are reported as conventional-minus-equivariant. Reconstruction differences used 10,000 paired cluster-bootstrap replicates at WSI level [19]. WSI diagnosis used 10,000 paired bootstrap replicates stratified by diagnosis. Tissue recognition used a paired WSI bootstrap stratified by observed tissue support. The five label-budget macro-F1 differences and the prespecified area under the log-budget learning curve (AULC) difference form a six-contrast family and use simultaneous 95% intervals. Patch-level standard deviations describe patch dispersion and do not replace WSI-level uncertainty. These procedures estimate sampling variation across the observed WSIs; they do not measure variation across new VAE or classifier training seeds.

IV. RESULTS

A. VAE Reconstruction

Both models reconstructed the major tissue structures and stain patterns in the predetermined validation panel shown in Fig. 3. The visual panel was fixed before the final comparison and was not selected from the test results.

The conventional VAE had lower pooled MAE and MSE and higher pooled PSNR and SSIM (Table III). Mean MAE was 0.062872 for the conventional model and 0.064230 for the equivariant model. The paired difference was -0.001358 , with a 95% interval of $[-0.002213, 0.000046]$. The interval included zero, so the primary analysis did not establish a reconstruction difference. MSE, PSNR and SSIM favored the conventional model descriptively.

B. Five-Class WSI Diagnosis

The equivariant representation obtained higher point estimates for macro-F1 and accuracy (Table IV). Macro-F1 was 0.483 versus 0.389, with a conventional-minus-equivariant difference of -0.095 and a 95% interval of $[-0.207, 0.050]$. Accuracy was 0.609 versus 0.522, with interval $[-0.217, 0.043]$. Neither interval established a difference. Balanced accuracy

TABLE III
RECONSTRUCTION ON 67,138 TEST PATCHES. VALUES ARE MEAN (POPULATION STANDARD DEVIATION).

Metric	Conventional	SO(2)
MAE ↓	0.0629 (0.0239)	0.0642 (0.0223)
MSE ↓	0.0080 (0.0074)	0.0083 (0.0067)
PSNR dB ↑	27.923 (2.721)	27.658 (2.595)
SSIM ↑	0.7575 (0.0842)	0.7436 (0.0862)

TABLE IV
OVERALL WSI DIAGNOSIS RESULTS ON 23 TEST SLIDES. DIFFERENCE INTERVALS ARE CONVENTIONAL-MINUS-SO(2).

Metric	Conventional	SO(2)	Difference 95% CI
Macro-F1	0.389	0.483	$[-0.207, 0.050]$
Balanced accuracy	0.510	0.503	$[-0.167, 0.180]$
Accuracy	0.522	0.609	$[-0.217, 0.043]$
Mean cross-entropy	1.323	0.993	$[0.135, 0.524]$

was nearly equal: 0.503 for the equivariant representation and 0.510 for the conventional representation.

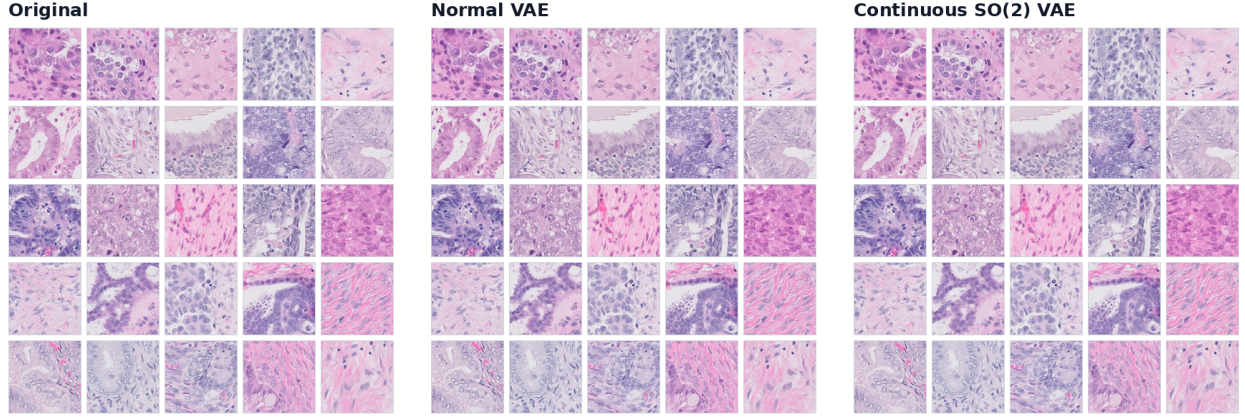
Mean cross-entropy was 0.993 for the equivariant representation and 1.323 for the conventional representation. Its difference interval, $[0.135, 0.524]$, favored the equivariant branch. Cross-entropy was a secondary endpoint and does not change the inconclusive primary macro-F1 result.

The classwise results show why the overall comparison remains uncertain (Table V). The conventional representation had zero F1 for EC and LGSC but classified both MC slides correctly. The equivariant representation improved EC and LGSC but failed on both MC slides. LGSC and MC each contained only two test WSIs, so a change of one prediction produces a large classwise difference.

C. Three-Class Tissue Recognition

The tissue results changed with the amount of labeled training data (Fig. 4 and Table VI). The equivariant representation had higher macro-F1 at 250, 500, 1,000 and 2,500 labels per class. At 500 labels per class, the macro-F1 values were 0.575 and 0.511, and the simultaneous interval for conventional-minus-equivariant was $[-0.121, -0.008]$. This was the only label budget whose simultaneous interval excluded zero.

Final reconstructions — same predetermined fixed validation 25



Ordered by frozen selector rank (five patches per source histotype); FP16 archives shown after $[-1,1] \rightarrow [0,1]$ clamping.

Fig. 3. Original patches and final reconstructions for the same 25 fixed VAE validation examples. This panel is descriptive. Quantitative reconstruction results use 67,138 patches from the 23 test WSIs.

TABLE V
CLASSWISE WSI F1 AND TEST SUPPORT.

Diagnosis	Support	Conventional F1	SO(2) F1
CC	5	0.667	0.750
EC	6	0.000	0.533
HGSC	8	0.706	0.800
LGSC	2	0.000	0.333
MC	2	0.571	0.000

TABLE VI
TISSUE MACRO-F1 ON 31,572 TEST PATCHES. SIMULTANEOUS INTERVALS ARE REPORTED FOR CONVENTIONAL-MINUS-SO(2).

Labels/class	Conventional	SO(2)	Simultaneous 95% CI
250	0.508	0.523	$[-0.072, 0.041]$
500	0.511	0.575	$[-0.121, -0.008]$
1,000	0.582	0.598	$[-0.072, 0.041]$
2,500	0.701	0.714	$[-0.069, 0.044]$
5,671	0.766	0.710	$[-0.001, 0.112]$

At 500 labels per class, the classwise F1 values for conventional/equivariant representations were 0.769/0.781 for tumor, 0.485/0.560 for stroma and 0.279/0.384 for necrosis. The supported difference at this budget was therefore not produced by tumor alone.

At the largest budget, the conventional representation obtained macro-F1 0.766 compared with 0.710 for the equivariant representation. The simultaneous interval included zero, with a lower bound of -0.001 . The corresponding classwise F1 values were 0.868/0.871 for tumor, 0.728/0.711 for stroma and 0.701/0.548 for necrosis. The normalized area under the log-budget learning curve was 0.615 for the conventional representation and 0.631 for the equivariant representation; its simultaneous difference interval included zero.

V. DISCUSSION

The first research question concerned reconstruction. The conventional VAE had lower pooled MAE and MSE and higher pooled PSNR and SSIM. The primary paired MAE interval included zero. The primary analysis therefore did not establish a reconstruction difference between the two implemented VAEs.

The second question concerned downstream classification. For WSI diagnosis, the equivariant representation obtained higher macro-F1 and accuracy, but the test set contained only 23 slides and the primary difference interval was wide. The per-class results were also unstable for LGSC and MC, which had two slides each. These values are point estimates and do not establish a WSI-level difference.

In tissue recognition, the equivariant representation had higher macro-F1 at the four smaller budgets. Only the 500-label interval excluded zero; the remaining budget intervals and the AULC interval included zero. At 5,671 labels per class, the conventional representation had higher macro-F1, largely because it had higher necrosis F1. These results do not establish a general increase in label efficiency.

The two models also differ in capacity and computation. The equivariant VAE has approximately 30% of the learned parameters of the conventional VAE, but its constrained kernel expansion requires more estimated multiply-accumulate operations. A result from this comparison may arise from equivariance, parameter count, numerical optimization, or their interaction. The experiment characterizes the two implemented representations; it does not separate these mechanisms.

Results differed by endpoint. Reconstruction means and the largest tissue budget favored the conventional VAE. Selected WSI and smaller-budget tissue point estimates favored the equivariant representation. The primary WSI interval and all but one tissue-budget interval included zero, so the experiment

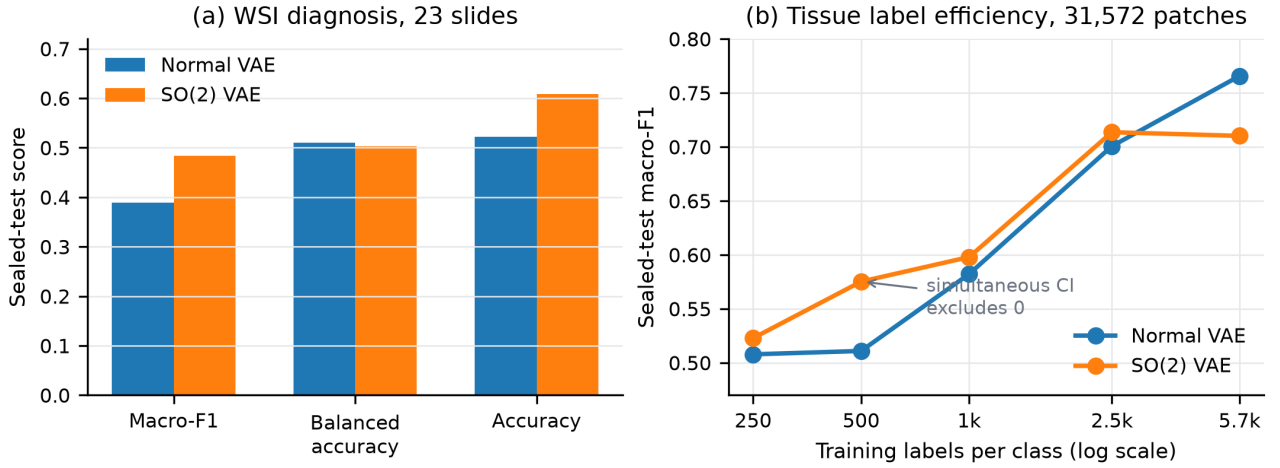


Fig. 4. Frozen-representation classification results. (a) Five-class WSI diagnosis on 23 test slides. (b) Tissue macro-F1 on 31,572 test patches under nested class-balanced training budgets. Only the 500-label-per-class tissue comparison has a simultaneous difference interval excluding zero.

does not establish a general ordering between the representations.

VI. LIMITATIONS

Each VAE and each downstream configuration was represented by one training trajectory. The bootstrap intervals quantify variation across the observed WSIs, not variation across new initializations, data orders or optimization runs. Repeated training would be required to estimate that component.

The WSI test contains 23 slides. LGSC and MC have two slides each, which limits classwise interpretation and produces wide macro-F1 intervals. Tissue recognition contains many patches but only 23 independent biological clusters; necrosis is present in five of them. Patch count therefore overstates the effective amount of independent test information.

Supplemental tissue masks are incomplete. The tissue experiment evaluates selected high-purity annotated regions and cannot be interpreted as exhaustive segmentation of complete WSIs. Results also come from one ovarian-cancer data source and have not been externally validated on another institution or organ.

The reconstruction aggregation and its bootstrap analysis were finalized after later test labels had been opened. Neither VAE checkpoint, architecture nor reconstruction output was changed at that stage, but the statistical analysis should be regarded as a finalized post-training evaluation rather than a fully preregistered endpoint.

Finally, the comparison does not match parameter count or computational work. A future experiment would require repeated seeds, a larger class-complete WSI test cohort and capacity-matched controls before attributing a classification difference specifically to continuous rotation equivariance.

VII. CONCLUSION

This study compared a conventional convolutional VAE and a continuous-SO(2) VAE trained on the same histopathology

patches and evaluated their frozen representations in reconstruction, WSI diagnosis and tissue recognition. The primary reconstruction analysis did not establish a difference between the two models. The equivariant representation obtained higher point estimates in WSI macro-F1 and accuracy, but the primary interval included zero. In tissue recognition, it had higher macro-F1 at four smaller label budgets, with an interval excluding zero only at 500 labels per class; the conventional representation had higher macro-F1 at the largest budget. Results varied by endpoint and did not establish a general advantage for either representation.

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